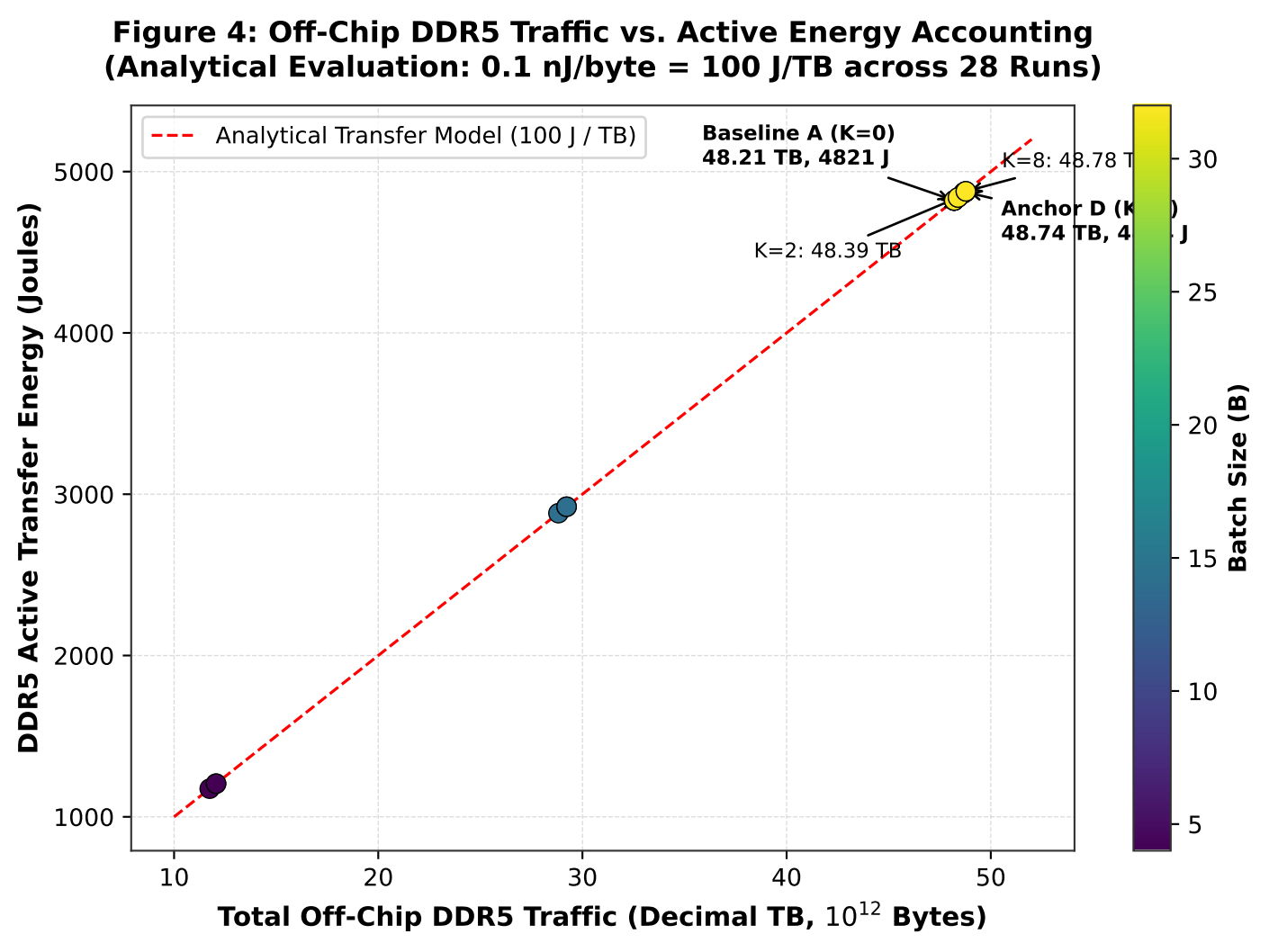
Figure 4: Off-Chip DDR5 Traffic vs. Active Energy Accounting
(Analytical Evaluation: 0.1 nJ/byte = 100 J/TB across 28 Runs)
Analytical Transfer Model (100 J / TB)
Baseline A (K=0)
48.21 TB, 4821 J
K=8: 48.78 T
Anchor D (K
48.74 TB, 4
J
K=2: 48.39 TB
DDR5 Active Transfer Energy (Joules)
5000
4000
3000
2000
1000
10
20
30
40
50
Total Off-Chip DDR5 Traffic (Decimal TB, 10^{12} Bytes)
Batch Size (B)
30
25
20
15
10
5